

Презентация

Лабораторная работа №4

Калашникова Д. В.

25 сентября 2026

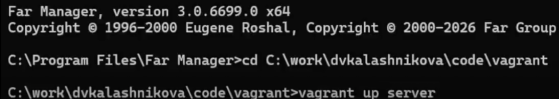
Российский университет дружбы народов, Москва, Россия

Информация

- Калашникова Дарья Викторовна
- Студент
- Российский университет дружбы народов
- 1132243108@pfur.ru

Приобретение практических навыков по установке и базовому конфигурированию HTTP-сервера Apache.

Запускаем виртуальную машину сервера через vagrant.



```
Far Manager, version 3.0.6699.0 x64  
Copyright © 1996-2000 Eugene Roshal, Copyright © 2000-2026 Far Group  
  
C:\Program Files\Far Manager>cd C:\work\dvkalashnikova\code\vagrant  
C:\work\dvkalashnikova\code\vagrant>vagrant up server
```

Рис. 1: Запуск сервера

Устанавливаем пакет `httpd` и сопутствующие утилиты для работы веб-сервера.

[illegible]

Рис. 2: Установка пакетов

Основной конфигурационный файл Apache — в нём задаются глобальные параметры сервера.

```

$ cd /etc/httpd/conf
$ vi httpd.conf
#
# To be able to use the functionality of a module which was built as a
# dynamic module (when you want to use a module that was built as
# a different module, add 'LoadModule' lines at this location so the
# dynamic module can be loaded at runtime.)
#
# DynamicSharedObject (DSO) support
#
# To be able to use the functionality of a module which was built as
# a DSO you have to place corresponding 'LoadModule' lines at this location
# so the module can be loaded at runtime.  Note the 'L' which means
# that a module must be loaded before any module which depends on it.
#
# DocumentRoot "path/to/html"
#
#
#
#
# Allow access to content within /usr/www:
#
<Directory "/usr/www">
    AllowOverride None
    # Allow open access:
    Require all granted
</Directory>
#
#
# Further relax access to the default document root:
<Directory "/usr/www/html">
    #
    # Possible values for the Options directive are "None", "All",
    # or any combination of:
    #   Indexes Includes FollowSymLinks SymLinks/OwnerMatch ExecCGI MultiViews
    #
    # Note that 'MultiViews' must be named 'explicitly' --- 'Options All'
    # doesn't give it to you.
    #
    # The Options directive is both complicated and important.  Please see
    # http://httpd.apache.org/docs/2.4/mod/core.html#options
    # for more information.
    #
    Options Indexes FollowSymLinks
#
#
#
#
# AllowOverride controls what directives may be placed in .htaccess files.
# It can be "All", "None", or any combination of the keywords:
#   Options FileInfo AuthConfig Limit
#
    AllowOverride None
#
# Controls who can get stuff from this server.
#
    Require all granted
</Directory>
#
#
#
#
# DirectoryIndex: sets the file that Apache will serve if a directory
# is requested.

```

Рис. 3: /etc/httpd/conf/httpd.conf

Файл с инструкциями для определения MIME-типа файла по его содержимому.

```
##
## This file contains magic binary patterns for guessing file types.
## The module is described in /usr/share/doc/httpd-2.4.18/magic.html
##
## The format is 4-5 columns:
## Column #1: byte number to begin checking from, ">" indicates continuation
## Column #2: type of data to match
## Column #3: contents of data to match
## Column #4: MIME type of result
## Column #5: MIME encoding of result (optional)
##
##-----
## Localstuff: file(s) magic for locally observed files
## Add any locally observed files here.
##
##-----
## end local stuff
##
##-----
## Java
##
0      short      0x4546        application/java
>12    short      0x4645
##
##-----
## audio: file(s) magic for sound formats
##
## From Jan Wiczlowski <jan@pft.vio.net>
##
## Sun/Net audio data
0      string      .and
>12    belong      1          audio/basic
>12    belong      2          audio/basic
>12    belong      3          audio/basic
>12    belong      4          audio/basic
>12    belong      5          audio/basic
>12    belong      6          audio/basic
>12    belong      7          audio/basic
>12    belong      23         audio/x-adpcm
##
## DEC systems (e.g. DECstation 5000) use a variant of the Sun/Net format
## that uses little-endian encoding and has a different magic number
## (0x00000000 in little-endian encoding).
0      belong      0x00000000
```

Рис. 4: /etc/httpd/conf/magic

Настройки автоматического отображения списка файлов в директории.

```
##
## /etc/httpd/conf.d/autoindex.conf
##
# Directives controlling the display of server-generated directory listings.
#
# Required modules: mod_auth_core, mod_authz_host,
#                  mod_autoindex, mod_alias
#
# To see the listing of a directory, the Options directive for the
# directory must include "Indexes", and the directory must not contain
# a file matching those listed in the DirectoryIndex directive.
#
#
# IndexOptions: Controls the appearance of server-generated directory
# listings.
#
IndexOptions FancyIndexing HTMLTable VersionSort

# We include the /icons/ alias for FancyIndexed directory listings.  If
# you do not use FancyIndexing, you may comment this out.
#
Alias /icons/ "/usr/share/httpd/icons/"

<Directory "/usr/share/httpd/icons">
    Options Indexes MultiViews FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>

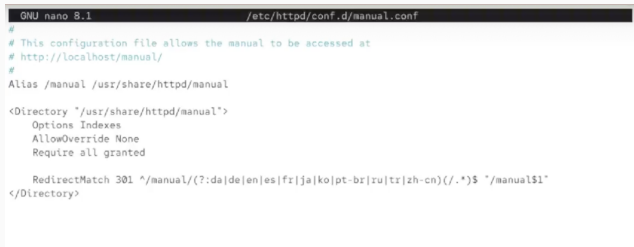
#
# AddIcon* directives tell the server which icon to show for different
# files or filename extensions.  These are only displayed for
# FancyIndexed directories.
#
AddIconByEncoding (CMP,/icons/compressed.gif) x-compress x-gzip

AddIconByType (TXT,/icons/text.gif) text/*
AddIconByType (IMG,/icons/image.gif) image/*
AddIconByType (SWF,/icons/movie2.gif) audio/*
AddIconByType (VID,/icons/movie.gif) video/*
AddIconByType (/icons/bomb.gif application/x-coredump)

AddIcon /icons/binary.gif .bin .exe
AddIcon /icons/binhex.gif .hxx
AddIcon /icons/tar.gif .tar
AddIcon /icons/world2.gif .wrl .wrl.gz .vml .vml.gz
AddIcon /icons/compressed.gif .Z .z .tgz .gz .zip
```

Рис. 5: /etc/httpd/conf.d/autoindex.conf

Настройки доступа к веб-странице с документацией Apache.



```
GNU nano 8.1 /etc/httpd/conf.d/manual.conf
#
# This configuration file allows the manual to be accessed at
# http://localhost/manual/
#
Alias /manual /usr/share/httpd/manual

<Directory "/usr/share/httpd/manual">
    Options Indexes
    AllowOverride None
    Require all granted

    RedirectMatch 301 ^/manual/(?:da|de|en|es|fr|ja|ko|pt-br|ru|tr|zh-cn)(/.*)$ "/manual$1"
</Directory>
```

Рис. 6: /etc/httpd/conf.d/manual.conf

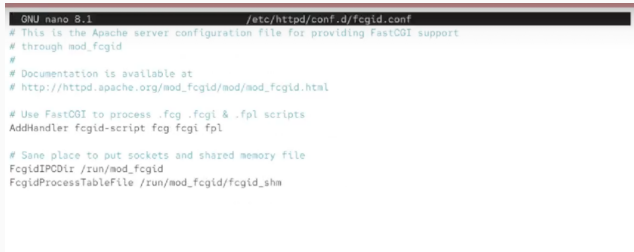
Настройки доступа к публичным веб-директориям пользователей системы.

```
GNU nano 8.1 /etc/httpd/conf.d/userdir.conf
#
# UserDir: The name of the directory that is appended onto a user's home
# directory if a ~user request is received.
#
# The path to the end user account 'public_html' directory must be
# accessible to the webserver userid. This usually means that ~userid
# must have permissions of 711. ~userid/public_html must have permissions
# of 755, and documents contained therein must be world-readable.
# Otherwise, the client will only receive a '403 Forbidden' message.
#
<IfModule mod_userdir.c>
#
# UserDir is disabled by default since it can confirm the presence
# of a username on the system (depending on home directory
# permissions).
#
UserDir disabled
#
# To enable requests to ~user/ to serve the user's public_html
# directory, remove the 'UserDir disabled' line above, and uncomment
# the following line instead:
#
#UserDir public_html
</IfModule>

#
# Control access to UserDir directories. The following is an example
# for a site where these directories are restricted to read-only.
#
<Directory "/home/*/public_html">
    AllowOverride FileInfo AuthConfig Limit Indexes
    Options MultiViews Indexes SymLinksIfOwnerMatch IncludesNoExec
    Require method GET POST OPTIONS
</Directory>
```

Рис. 7: /etc/httpd/conf.d/userdir.conf

Настройки модуля mod_fcgid для запуска скриптов через FastCGI.

A screenshot of a terminal window showing the contents of the file /etc/httpd/conf.d/fcgid.conf. The terminal title bar indicates 'GNU nano 8.1' and the file path. The file contains configuration comments and directives for the mod_fcgid module, including documentation links, script types, and socket/shm file locations.

```
GNU nano 8.1 /etc/httpd/conf.d/fcgid.conf
# This is the Apache server configuration file for providing FastCGI support
# through mod_fcgid
#
# Documentation is available at
# http://httpd.apache.org/mod_fcgid/mod/mod_fcgid.html
#
# Use FastCGI to process .fcg .fcgi & .fpl scripts
AddHandler fcgid-script fcg fcgi fpl
#
# Sane place to put sockets and shared memory file
FcgidIPCDir /run/mod_fcgid
FcgidProcessTableFile /run/mod_fcgid/fcgid_shm
```

Рис. 8: /etc/httpd/conf.d/fcgid.conf

Настройки поддержки шифрования SSL/TLS.

```
##
##
## When we also provide SSL, we have to listen to the
## standard HTTPS port in addition.
##
Listen 443 https

##
## SSL Global Context
##
## All SSL configuration in this context applies both to
## the main server and all SSL-enabled virtual hosts.
##

# Pass Phrase Dialog:
# Configure the pass phrase gathering process.
# The filtering dialog program (builtin) is a internal
# temporal dialog) has to provide the pass phrase on stdout.
SSLPassPhraseDialog exec:/usr/libexec/httpd-ssl-pass-dialog

# Inter-Process Session Cache:
# Configure the SSL Session Cache: First the mechanism
# to use and second the expiring timeout (in seconds).
SSLSessionCache         shmcb:/run/httpd/sslcache(512000)
SSLSessionCacheTimeout  300

#
# Use the "SSLCryptoDevice" to enable any supported hardware
# accelerators. Use "openssl engine -a" to list supported
# engine names. NOTE: If you enable an accelerator and the
# server does not start, consult the error logs and ensure
# your accelerator is functioning properly.
#
SSLCryptoDevice builtin
#SSLCryptoDevice ubsec

##
## SSL Virtual Host Context
##

<VirtualHost *:443>

# General setup for the virtual host, inherited from global configuration
#DocumentRoot "/var/www/html"
#ServerName www.example.com:443
```

Рис. 9: /etc/httpd/conf.d/ssl.conf

Добавляем службу http в фаервол, чтобы клиент мог обращаться к веб-серверу.

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
bash: firewall-cmd: command not found...
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns ssh
[root@server.dvkalashnikova.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quit dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-clien
t ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 ip
erf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadm1n kdeconnect kerberos kibana klogi
n kpssd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controll
er-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kub
e-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmr l
lmnr-client llmr-tcp llmr-udp managesieve matrix ndns memcache minecraft minidna mdp mongodb mosh mou
ntd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-
dashboard nfs nfs3 nmap-0183 nripe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-
vncconsole plex pncd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius radsec rdp red
is redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane set
tlers-history-collection sip sips slimevr sip smtp smtp-submission smtps snmp snmptls snmptls-trap snmptr
ap socks spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming stellar
is stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-rel
ay synergy sysconlan syslog syslog-ng syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks tran
smision-client turn turns upnp-client vdm vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-
discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wssd wssd-http wsmn ws
mans xdmpc xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server za
bbix-trapper zabbix-web-service zero-k zerotier
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http --permanent
bash: firewall-cmd: command not found...
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=http --permanent
success
```

Рис. 10: Настройка firewall

Запускаем journalctl с ключом -f для отслеживания логов в реальном времени.

```
Subject: Process 18242 (VBoxClient) dumped core
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support
Documentation: man:core(5)

Process 18242 (VBoxClient) crashed and dumped core.

This usually indicates a programming error in the crashing program and
should be reported to its vendor as a bug.
Sep 24 15:57:59 server.dvkalashnikova.net systemd[1]: systemd-coredump@202-18246-0.service: Deactiva
ccessfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-coredump@202-18246-0.service has successfully entered the 'dead' state.
```

Рис. 11: journalctl

Запускаем службу httpd и включаем её автозагрузку.

```
[root@server.dvkalashnikova.net ~]# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
[root@server.dvkalashnikova.net ~]# systemctl start httpd
```

Рис. 12: запуск службы httpd

Лог об успешном запуске

В журнале видим, что запуск httpd прошёл успешно.

```
should be reported to its vendor as a bug.
Sep 24 15:58:38 server.dvkalashnikova.net systemd[1]: systemd-coredump@208-18711-0.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-coredump@208-18711-0.service has successfully entered the 'dead' state.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit serial-getty@ttyS0.service has successfully entered the 'dead' state.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 142.
Subject: Automatic restarting of a unit has been scheduled
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

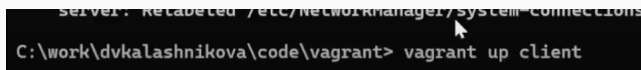
Automatic restarting of the unit serial-getty@ttyS0.service has been scheduled, as the result for the configured Restart= setting for the unit.
Sep 24 15:58:39 server.dvkalashnikova.net systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Subject: A start job for unit serial-getty@ttyS0.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit serial-getty@ttyS0.service has finished successfully.

The job identifier is 32606.
```

Рис. 13: Лог об успешном запуске

Запускаем виртуальную машину клиента через vagrant.



```
Server: Retabeted /etc/networkmanager/system-connections  
C:\work\dvkalashnikova\code\vagrant> vagrant up client
```

Рис. 14: Запуск клиента

Переходим в браузере клиента по адресу 192.168.1.1 — видим страницу по умолчанию.

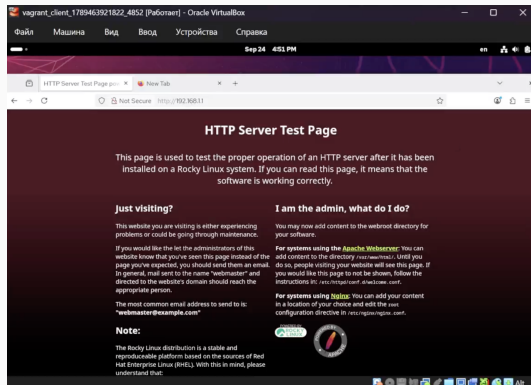


Рис. 15: Страница по умолчанию

Смотрим лог ошибок веб-сервера — он пуст, ошибок нет.

```
[dvkashnikova@server.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkashnikova:
[root@server.dvkalashnikova.net ~]# tail -f /var/log/httpd/error_log
[Thu Sep 24 15:58:35.743830 2026] [suexec:notice] [pid 18525:tid 18525] AH01232: suEXEC mechanism enabled
(wrapper: /usr/sbin/suexec)
[Thu Sep 24 15:58:35.976442 2026] [lbmethod_heartbeat:notice] [pid 18525:tid 18525] AH02282: No slotmem f
rom mod_heartbeat
[Thu Sep 24 15:58:36.001799 2026] [systemd:notice] [pid 18525:tid 18525] SELinux policy enabled; httpd ru
nning as context system_u:system_r:httpd_t:s0
[Thu Sep 24 15:58:36.046646 2026] [mpm_event:notice] [pid 18525:tid 18525] AH00489: Apache/2.4.63 (Rocky
Linux) OpenSSL/3.5.5 mod_fcgid/2.3.9 configured -- resuming normal operations
[Thu Sep 24 15:58:36.046721 2026] [core:notice] [pid 18525:tid 18525] AH00094: Command line: '/usr/sbin/h
ttpd -D FOREGROUND'
```

Рис. 16: /var/log/httpd/error_log

Смотрим лог доступа — в нём фиксируется обращение клиента к серверу.

```
[dvkashnikova@server.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkashnikova:
[root@server.dvkalashnikova.net ~]# tail -f /var/log/httpd/access_log
192.168.1.2 - - [24/Sep/2026:16:48:42 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0 (X
11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.2 - - [24/Sep/2026:16:48:43 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15443
"http://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/1
40.0"
192.168.1.2 - - [24/Sep/2026:16:48:43 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "http:
//192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.2 - - [24/Sep/2026:16:48:44 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://1
92.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
█
```

Рис. 17: /var/log/httpd/access_log

Останавливаем DNS-службу, чтобы заменить файлы зон.

```
[root@server.dvkalashnikova.net ~]# systemctl stop named  
[root@server.dvkalashnikova.net ~]#
```

Рис. 18: Остановка службы named

Добавляем в прямую зону запись о новом http-сервере www.

```
GNU nano 8.1 /var/named/master/fz/dvkalashnikova.net Modified
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                2026091501      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )   ; minimum

      NS      @
      A       192.168.1.1
      AAAA    ::1
$ORIGIN dvkalashnikova.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
www A 192.168.1.1
```

Рис. 19: Файл зоны /var/named/master/fz/dvkalashnikova

Файл зоны /var/named/master/rz/192.168.1

Аналогичную запись добавляем в обратную зону.

```
GNU nano 8.1 /var/named/master/rz/192.168.1 Mo
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                2026091501      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )   ; minimum

      NS      @
      A       192.168.1.1
      AAAA    ::1
      PTR     server.dvkalashnikova.net
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.dvkalashnikova.net.
1 PTR ns.dvkalashnikova.net.
1 PTR dhcp.dvkalashnikova.net.
1 PTR www.dvkalashnikova.net.
```

Рис. 20: Файл зоны /var/named/master/rz/192.168.1

Удаляем журналы из папок прямой и обратной зон, чтобы применить изменения.

```
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# nano /var/named/master/fz/192.168.1
[root@server.dvkalashnikova.net ~]# nano /var/named/master/rz/192.168.1
[root@server.dvkalashnikova.net ~]# ls /var/named/master/fz/
dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# ls /var/named/master/fz/
dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# ls /var/named/master/rz/
192.168.1
```

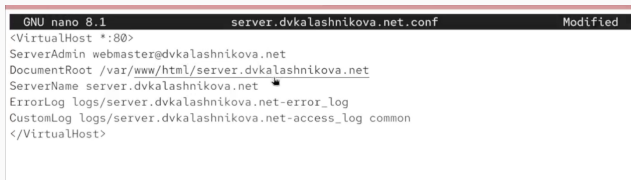
Рис. 21: Удаление журналов

Создаём файлы конфигурации для двух страниц — server и www.

```
[root@server.dvkalashnikova.net ~]# systemctl start named  
[r File display r.dvkalashnikova.net ~]# cd /etc/httpd/conf.d  
[root@server.dvkalashnikova.net conf.d]# ouch server.dvkalashnikova.net.conf  
bash: ouch: command not found...  
[root@server.dvkalashnikova.net conf.d]# touch server.dvkalashnikova.net.conf  
[root@server.dvkalashnikova.net conf.d]# touch www.dvkalashnikova.net.conf  
[root@server.dvkalashnikova.net conf.d]# nano server.dvkalashnikova.net.conf
```

Рис. 22: Создание конфигурационных файлов

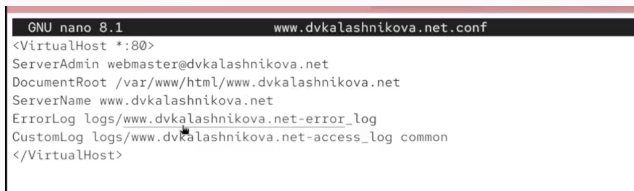
Помещаем в первый файл настройки виртуального хоста server.dvkalashnikova.net.



```
GNU nano 8.1          server.dvkalashnikova.net.conf      Modified
<VirtualHost *:80>
ServerAdmin webmaster@dvkalashnikova.net
DocumentRoot /var/www/html/server.dvkalashnikova.net
ServerName server.dvkalashnikova.net
ErrorLog logs/server.dvkalashnikova.net-error_log
CustomLog logs/server.dvkalashnikova.net-access_log common
</VirtualHost>
```

Рис. 23: server.dvkalashnikova.net

Во второй файл помещаем настройки виртуального хоста `www.dvkalashnikova.net`.

A screenshot of a terminal window with a dark background. The title bar at the top shows 'GNU nano 8.1' on the left and 'www.dvkalashnikova.net.conf' on the right. The terminal content shows the configuration of a VirtualHost in XML format. The lines are: <VirtualHost *:80>, ServerAdmin webmaster@dvkalashnikova.net, DocumentRoot /var/www/html/www.dvkalashnikova.net, ServerName www.dvkalashnikova.net, ErrorLog logs/www.dvkalashnikova.net-error_log, CustomLog logs/www.dvkalashnikova.net-access_log common, and </VirtualHost>. A cursor is visible at the end of the CustomLog line.

```
GNU nano 8.1                               www.dvkalashnikova.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@dvkalashnikova.net
DocumentRoot /var/www/html/www.dvkalashnikova.net
ServerName www.dvkalashnikova.net
ErrorLog logs/www.dvkalashnikova.net-error_log
CustomLog logs/www.dvkalashnikova.net-access_log common
</VirtualHost>
```

Рис. 24: `www.dvkalashnikova.net`

Создаём папки сайтов и кладём в них простые приветственные страницы index.html.

```
[root@server.dvkalashnikova.net conf.d]# cd /var/www/html
[root@server.dvkalashnikova.net html]# mkdir server.dvkalashnikova.net
[root@server.dvkalashnikova.net html]# cd /var/www/html/server.dvkalashnikova.net
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# touch index.html
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# echo "Welcome to the server.dvkalashnikova.net server." > index.html
[root@server.dvkalashnikova.net server.dvkalashnikova.net]# cd /var/www/html
[root@server.dvkalashnikova.net html]# mkdir www.dvkalashnikova.net
[root@server.dvkalashnikova.net html]# cd /var/www/html/www.dvkalashnikova.net
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# touch index.html
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# echo "Welcome to the www.dvkalashnikova.net server." > index.html
```

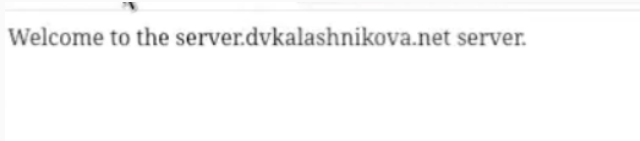
Рис. 25: Файлы index.html

Обновляем метки SELinux и перезапускаем службу httpd.

```
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# chown -R apache:apache /var/www
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /etc
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /var/named
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# restorecon -vR /var/www
[root@server.dvkalashnikova.net www.dvkalashnikova.net]# systemctl restart httpd
```

Рис. 26: Обновление меток и перезапуск службы

С клиента переходим по адресу `server.dvkalashnikova.net` — видим приветственную страницу.

A screenshot of a web browser window. The address bar shows the URL 'server.dvkalashnikova.net'. The main content area of the browser displays the text 'Welcome to the server.dvkalashnikova.net server.' in a monospaced font.

Welcome to the server.dvkalashnikova.net server.

Рис. 27: server.dvkalashnikova.net

Обновление конфигурации vagrant

Переносим готовые конфиги в папку provision/server и создаём скрипт http.sh.

```
[root@server.dvkalashnikova.net server]# cd /vagrant/provision/server/dns/
[root@server.dvkalashnikova.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/
named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/dvkalashnikova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol
error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Pro
tocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost':
Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback':
Protocol error
[root@server.dvkalashnikova.net dns]# cd /vagrant/provision/server
[root@server.dvkalashnikova.net server]# touch http.sh
[root@server.dvkalashnikova.net server]# chmod +x http.sh
```

Рис. 28: Обновление конфигурации vagrant

В скрипте http.sh описываем алгоритм автоматической настройки http-сервера.

```
GNU nano 8.1                               http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

Рис. 29: Скрипт http.sh

В Vagrantfile добавляем запуск скрипта http.sh при поднятии сервера.

```
server.vm.provision "server dummy", type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"  
server.vm.provision "server forward", type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forward.sh"  
server.vm.provision "server dns", type: "shell", preserve_order: true, path: "provision/server/dns.sh"  
server.vm.provision "server dhcp", type: "shell", preserve_order: true, path: "provision/server/dhcp.sh"  
server.vm.provision "server http", type: "shell", preserve_order: true, path: "provision/server/http.sh"  
end
```

Рис. 30: Vagrantfile

В результате выполнения лабораторной работы были получены навыки работы и настройки http сервера